Dr. Manuel Meyer | Curriculum vitæ

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Personal Details

- Languages: German (native), English (fluent), Swedish (proficient), Danish (advanced), French (basic), Spanish (basic)
- Identifiers: ORCID:0000-0002-0738-7581, INSPIRE HEP: Manuel.Meyer.1, Scopus Researcher ID: 55463373300, Web of Science: E-2697-2016, Google Scholar

Research Interests

My interests lie in the field of astroparticle physics and experimental searches for dark matter, in particular in the form of axions and axion-like particles. My research has provided stringent limits on the coupling between photons and axion-like particles using astrophysical observations. I also participate in the characterization and commissioning of the transition edge sensor to be used in the ALPS II experiment and lead the development of machine-learning-based data analysis.

Experience

Since 2023 **Associate Professor for Experimental Physics** Odense, Denmark

- Center for Cosmology and Particle Physics Phenomenology (CP3), University of Southern Denmark
- Group website

2021-2022 **Research group leader** Hamburg, Germany

- Institute for Experimental Physics, University of Hamburg
- Parental leave: July 2021, March-July 2022

2019 - 2021 ***Marie Skłodowska-Curie* Research Fellow** Erlangen, Germany

- Erlangen Center for Astroparticle Physics, University Erlangen-Nuremberg. PI: Prof. Stefan Funk

2019 ***Feodor Lynen* Research Return Fellow** Hamburg, Germany

- Deutsches Elektron Synchrotron (DESY). PI: Dr. Axel Lindner

2017 - 2019 ***Feodor Lynen* Research Fellow** Stanford, CA, USA

- Kavli Institute for Particle Astrophysics and Cosmology, Stanford University. PI: Prof. Roger Blandford

2013 - 2016 **Post doctoral researcher** Stockholm, Sweden

- Oskar Klein Centre for Cosmoparticle Physics, Stockholm University. PI: Prof. Jan Conrad

Education

2010 - 2013 **Ph.D., Physics, University of Hamburg** Hamburg, Germany

- Ph.D. thesis: The opacity of the Universe for high and very high energy γ-rays.
  Grade: "Magna Cum Laude" – with Honors
- Advisor: Prof. Dieter Horns

2004 - 2010 **Diploma in Physics, University of Hamburg** Hamburg, Germany

- Diploma thesis: Spectral Modeling of the Crab Nebula and Search for oscillatory features in X-ray data induced by hidden Photons. *Grade: "Excellent, with Honors"*
- Advisor: Prof. Dieter Horns

Selected Grants

2025 **DFF Project 1 Grant** Odense, Denmark

- Research grant awarded by Denmark's Independent Research Fund (DFF).
- Amount of funding: 3,168,000 DKK (ca. 424,000€)

2025 **Thrige Foundation Grant** Odense, Denmark

- Small research grant awarded by the private Thrige Foundation.
- Amount of funding: 100,000 DKK (ca. 13,400€)

2021 **DFG Research Unit** Bonn, Germany

- Associated member of a research group funded by the German Research Foundation (DFG).
- Amount of funding: 3,600,000€

2020 **ERC Starting Grant** Brussels, Belgium

- Research grant for early career scientists awarded by European Research Council (ERC).
- Amount of funding: 1,441,000€

2019 ***Marie Skłodowska-Curie* Research Fellowship** Brussels, Belgium

- Marie Skłodowska-Curie Actions Research Fellowship awarded by the ERC.
- Amount of funding: 175,000€

2016 ***Feodor Lynen* Research Fellowship** Bonn, Germany

- Sponsorship for own research position from the German Humboldt Foundation.
- Amount of funding: 126,000€

Awards

2017 **Marie Skłodowska-Curie Actions Seal of Excellence** Brussels, Belgium

- Award for proposal submitted under the Horizon 2020 Marie Skłodowska-Curie actions

2010 ***Otto Stern* Prize** Hamburg, Germany

- Prize for best diploma thesis in physics of the 2010 summer term at the University of Hamburg

Collaboration Memberships

since 2021 **Any Light Particle Search II (ALPS II) Collaboration**

- Leading the machine-learning-based data analysis development and system detection efficiency measurement for the TES detector; member of the executive board

since 2013 ***Fermi* Gamma Ray Space Telescope Collaboration**

- Former coordinator of the dark matter and new physics working group

since 2012 **Cherenkov Telescope Array Observatory (CTAO) Consortium**

- Member of the Speakers and Publications Office, former convener of the Dark Matter and Exotic Physics working group

since 2012 **High Energy Stereoscopic System (H.E.S.S.) Collaboration**

- Member of steering committee, convener of the physics beyond the standard model working group, former member of the multi-wavelength board

Institutional and Organizational Boards

since 2023 **Institutional committees at SDU** Odense, Denmark

- Member of evaluation committee for 2 professor positions including being head of external search committee; head of astrophysics education at SDU; completed training for laser safety officers

since 2024 **SDU Quantum Hub** Odense, Denmark

- Member of the steering committee

since 2023 **European COST Action CA21106 "Cosmic WISPers"**

- Danish country representative in management committee

since 2023 **Quantum Universe Excellence Cluster** Hamburg, Germany

- Representative of the young science group leaders in the steering committee

Selected Publications

51 publications submitted or published in peer-reviewed journals with major contribution (19 as corresponding author). Co-author of more than 65 publications of the H.E.S.S. collaboration, 25 publications of the *Fermi*-LAT collaboration, 28 conference proceedings, and 5 white papers. According to NASA ADS, the publications have in total more than 13,700 citations with an h index of 53. A publication list including all collaboration papers can be found on ordid.org/orcid.org/0000-0002-0738-7581.

1. S. Porras-Bedmar, M. **Meyer**, and D. Horns, *Novel bounds on decaying axionlike particle dark matter from the cosmic background*, Phys. Rev. D, Vol. 110, No. 10, 103501, p. 103501, 2024, arXiv: 2407.10618 [astro-ph.CO]
2. M. **Meyer** et al., *A First Application of Machine and Deep Learning for Background Rejection in the ALPS II TES Detector*, Annalen der Physik, Vol. 536, No. 1, 2200545 2024, arXiv: 2304.08406 [hep-ex]
3. J. Rubiera Gimeno et al. (including M. **Meyer**), *The TES detector of the ALPS II experiment*, Nuclear Instruments and Methods in Physics Research A, Vol. 1046, 167588, p. 167588, 2023
4. J. Biteau and M. **Meyer**, *Gamma-Ray Cosmology and Tests of Fundamental Physics*, Galaxies, Vol. 10, No. 2, p. 39, 2022, arXiv: 2202.00523 [astro-ph.CO]
5. M. **Meyer** and T. Petrushevska, *Search for Axionlike-Particle-Induced Prompt γ-Ray Emission from Extragalactic Core-Collapse Supernovae with the Fermi Large Area Telescope*, Phys. Rev. Lett. Vol. 124, No. 23, 231101, p. 231101, 2020, arXiv: 2006.06722 [astro-ph.HE]

Conference Contributions

Since 2013, I have given 22 invited talks at conferences, given lectures in 3 international summer schools, and presented my research at 10 invited university colloquia.

Selected Invited Presentations

2024	2nd General Meeting of COST Action COSMIC WISPers (plenary talk)	Turkey
2024	TeV Particle Astrophysics Conference (plenary talk)	USA
2024	15th International Workshop on the Identification of Dark Matter (plenary talk)	Italy
2024	Generation, evolution, and observations of cosmological magnetic fields Workshop	Switzerland
2023	Danish Physical Society meeting (plenary talk)	Denmark

Supervision & Mentoring

Winter 2025	Current group: 1 PI, 3 post docs, 1 PhD student, 1 master student
Since 2023	Mentor for the SDU Physics Mentoring program (currently 2 mentees)
Since 2021	Mentor of the Fermi-LAT mentoring program (2 mentees in past cycles)
Since 2018	Main supervisor of 1 post doc: Gulden Othman (2021-2024) and 1 PhD student: Rahul Cecil (2021-2025); co-supervision of 3 PhD students: J. Davies (2020-2022), M. Crnogorčević (2018-2023), A. Yusafzai (2020-2022).
Since 2014	Main supervisor of 5 master students and 8 bachelor students; co-supervisor of 4 master students and 1 bachelor student

Teaching and Formal Training in Teaching

Courses and Classes taught

Since 2024	**Astroparticle Physics**	Odense, Denmark

- 10 ECTS lecture and exercise classes for master students.

Since 2024	**Electrodynamics**	Odense, Denmark

- 5 ECTS lecture and exercise classes for undergraduate students.

Since 2023	**Research Activities in Experimental Physics**	Odense, Denmark

- 5 ECTS course project work on experimental physics and data analysis for small groups of graduate students.

2023 - 2024	**Introduction to Quantum Mechanics**	Odense, Denmark

- 5 ECTS lecture and exercise classes for undergraduate students

2021 - 2022	**Particle and Astroparticle Physics**	Hamburg, Germany

- Supervising and evaluating seminars prepared by students.

2020 - 2022	**Gamma-ray Telescope in the Classroom**	Erlangen, Germany

- Lectures, exercise classes, and observation nights for bachelors and masters students.

2019 - 2020	**Data Analysis II and Introduction to Machine Learning**	Erlangen, Germany

- Lectures and exercise classes given for masters students.

2018	**Lectures for pre-collegiate students**	Stanford, CA, USA

- Lectures given for high school students during particle physics and cosmology summer courses.

2011-2012	**Lab assistant for undergraduate physics students**	Hamburg, Germany

- Supervision of small groups of students for week-long labs. Included teaching, supervision, feedback and grading.

Training

2018	**Stanford Scientific Teaching Summer Institute**	Stanford, CA, USA

- Three day workshop on science-based teaching, inclusion, and equity.

2017	**Introduction to Undergraduate STEM Teaching**	Stanford, CA, USA

- Online course offered by the Center for the Integration of Research Teaching and Learning, stemteachingcourse.org. Passed with distinction.

Outreach

9 outreach talks for general public and high schools students, **8 articles for press releases, blogs, and newspapers**. Examples can be found here, here, here, here, here, here, here, here, here, and here. Also worked as a advisor for high school students at the DESY school labs (2008-2009).

Computer skills

- Operating systems: Mac OS, Ubuntu, Windows
- Programming languages: python, Root, C++, C
- Scientific Computing: numpy, scipy, astropy, scikit-learn, iminuit, emcee, tensorflow & keras, matplotlib, hdf5, pandas, git
- Developed Software: gammaALPs, ebltable

◗ Data analysis	*Fermi*-LAT (FERMIPY and *Fermi Tools*); H.E.S.S. and CTA (HAP and GAMMAPY); *Swift* XRT, NuSTAR, and XMM-Newton (pipelines and XSPEC); ALMA (CASA), ALPS II (TES)
◗ Document preparation	LaTeX, Keynote, OpenOffice, Microsoft Office

◗◗◗ Reviewing

◗ Member of scientific organizing committees	Phystat gamma conference 2022, International Cosmic Ray Conference (ICRC) 2021; "Recontres du Vietnam: Very High Energy Phenomena in the Universe" 2018 conference; "The puzzle of dark matter 2018" conference
◗ Journal reviewer	Nature Astronomy, Physical Review Letters, Physical Review D, Astrophysical Journal (ApJ), the Journal of Cosmology and Astroparticle Physics (JCAP), Physics Letters B, Monthly Notes of the Royal Astronomical Society (MNRAS), European Physical Journal C